

Overview of Autism Spectrum Disorder (ASD)

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Abstract:

Autism spectrum disorder (ASD) is a neurodevelopmental disorders that manifest during the early developmental period of childhood. Autism spectrum disorder (ASD) is a neurodevelopmental disorders defined by impairments in social interaction and communication, in combination with restricted and stereotyped patterns of behaviour and/or interest. There is, however, nosed individuals, including a range of cognitive ability that spans diverse range of symptom presentation and severity among diagnostic IQ scores representative of profound intellectual disability through to average and well above-average levels of intelligence. As a result, in less severe cases, symptoms may not become fully apparent until limited abilities. ASD is a new diagnostic label, chosen by the later stages of life when social demands surpass the individual's h American Psychiatric Association the fifth edition of their Diagnostic and Statistic Manual (DSM-5) [1]. This label was designed to provide a singular diagnosis to individuals who would have previously been categorized by one of the pervasive developmental disorders (PDDs), ASD defined in the International Classification of Diseases (ICD-11) as by persistent deficits in social communication and interaction and restricted, repetitive, and inflexible patterns of behavior, interests, or activities, with onset during the developmental period. The definition includes the importance of significant impairment in functioning and

specifies the inclusion of unusual sensory responses and the distinction between ASD with and without intellectual disability. [2.3.4].

Keywords: Autism, intellectual disability, communication, pervasive developmental disorders, intelligence, cognitive ability.

Discussion:

History of autism spectrum disorders

Leo Kanner, an Austrian-born child psychiatrist, working at Johns Hopkins University, is credited with the first clinical portrayal of ASD in his 1943 paper entitled Autistic disturbances of affective contact [5]. In this paper, Kanner details the behaviour of 11 children displaying a pattern of symptoms which included intense resistance to change, insistence on 'aloneness, and impairments in both communication skills and social interaction. Meanwhile, in 1944, independent of the work being conducted by Kanner, Hans Asperger (a psychiatrist working in Austria) produced his own clinical accounts of four cases of what he referred to as autistic personality disorder in his paper Die Autistischen Psychopathen im Kindesalter [6]. Similar to what was being described by Kanner, Asperger notes impairments in social and emotional relationships, abnormalities in speech and non-verbal communication, and restricted, but intense, focus on special interests. The primary difference between their accounts is that Asperger additionally discusses the presence of extraordinary skills in particular cognitive domains (for example, mathematics) in the children he was observing. Unfortunately, due to the fact that his work was published in German, Asperger's clinical findings did not become widely disseminated in English-speaking regions until nearly 40 years later.

The first widely utilized label for this disorder ('early infantile autism') was provided in a 1956 paper co-authored by Kanner and Leon Eisenberg [7]. In this paper, the authors proposed that two core features define early infantile autism: 'obsessive insistence on same-ness' and 'extreme self-isolation. Kanner had observed that parents of the patients visiting his clinic presented with similar social disin-terest and emotionally distant characteristics to those expressed by their children. Based on these observations, Eisenberg and Kanner proposed that emotional deprivation, as a result of this distant and 'cold' parenting, could play a role in the development of early in-fantile autism. However, they also went on to express that emotional deprivation could not solely account for the development of early infantile autism, as it was their belief that this disorder was a state that was present at birth and, as such, was likely to be the result of a combination of biological, psychological, and social factors. Unfortunately, it was their declaration regarding the potential. role of socially detached, 'cold parenting tion initially, and eventually resulted in what becomes known as the 'refrigerator mother' theory of autism. In the 1960s, this theory was furthered in publications by Bruno Bettelheim [8, 9], who, for many years, was considered as one of the leading experts in the field of autism, resulting in the 'refrigerator mother' theory becoming widely disseminated and accepted by many in the lay population, as well as in medical fields. Beginning in the late 1960s, and more significantly in the 1970s, research in the field of autism moved away from the clinical descriptions and/or theory presentations that previously dominated the landscape of autism literature, and began applying more scientif-ically stringent methodologies to the study of this disorder. Works produced during this period presented serious challenges to the psychogenic theories of autism (for example, the

'refrigerator mother' theory), by providing empirically supported evidence for a biological and neurological basis of the disorder [10, 11]. In addition, evidence was produced against the then-held position that autism was a form of childhood schizophrenia, with comparative studies clearly distinguishing the two disorders by their associated symptoms, age of onset, and family history [12, 13]. Following these scientific advances, 'infantile autism' was introduced as a formal diagnostic category by the American Psychiatric Association the 1980 release DSM-III [14].

In 1981 [15], Lorna Wing introduces the findings of Hans Asperger to the English-speaking world and describes 'Asperger's syndrome, a milder manifestation of the clinical symptoms of infantile autism, often paired with strong cognitive abilities in fields such as mathematics, and excluding the significant language development delays associated with autism. In DSM-IV [2] and DSM-IV-TR [3], the association between this Asperger's syndrome and autism (then labelled as 'autistic disorder) was officially recognized by the American Psychiatric Association, with the decision to expand the diagnostic category of PDDs to include five disorders that were thought to be related to, or share, many of the same features of one another: autistic disorder, Asperger's syndrome, Rett's syndrome, childhood disintegrative disorder, and PDD-NOS. Following this expansion, this group of disorders (with the exception of Rett's syndrome) began to be discussed under the umbrella term of 'autism spectrum disorders' in much of the autism literature. In 2013, the fifth, and most recent, edition of the DSM (DSM-5) [1] was released and was met with a great deal of controversy regarding its alterations to the diagnostic categorizations of autism and its related disorders. In DSM-5, the previous distinct categories of autistic disorder - childhood disintegrative disorder. Asperger's syndrome,

and PDD-NOS have been conceptualized as a singular 'autism spectrum disorder' and are defined by more rigid diagnostic criteria. This change was made in an effort to provide a more unified and rigorous definition of what autism is; however, it was met with a number of concerns and criticisms from medical professionals, as well as autism advocacy groups (see Concerns and criticisms of DSM-5 classification of autism spectrum disorder, p. 242, which contains further discussion on this topic), including negative impacts on the sense of community and identity that had been gained through specific diagnostic labels and potential issues with impaired.

Prevalence estimates.

Children not meeting the more stringent diagnostic criteria and losing access to critical services as a result. To address the former of these concerns, while Asperger's syndrome and PDD-NOS are no longer diagnostic options for clinicians within the framework of DSM-5, it was decided that a 'grandfather' clause would be introduced, allowing these labels to remain in individuals who received their diagnosis prior to this change. Meanwhile, to address the concerns over diagnostic sensitivity, DSM-5 introduced a new condition called social communication disorder, intended to provide a label for children without the repetitive and stereotyped behaviour. The components of ASD but who do express significant social/communicative impairments.

The first study to present data regarding an estimated prevalence rate for autism was performed by Victor Lotter in 1966¹⁶. To provide this estimate, Lotter screened the entire population of 8- to 10-year-old children (in total) for behavioural markers of early infantile autism, year-old children from a county in the south east of England (78,000

as then defined by Eisenberg and Kanner [6]. The findings of this screening resulted in an estimate that autism was present in 4.5 of every 10,000 children. From this point until as late as the 1990s, autism was considered to be a rare condition, with subsequent published prevalence reports suggesting rates of 2-5 per 10,000 children [17, 18]. Following the inclusion of 'Infantile autism' in DSM-III in 1980 [14], and the subsequent expansion of diagnostic criteria in logical studies began documenting dramatic increases in 1987 with release of the revised DSM-III-R [19], reported epidemiological rates of prevalence for this disorder [20-24, 29].

Beginning in the early 2000s, the Center for Disease Control and Prevention (CDC) began collecting large data samples of 8-year old children from across the United States and producing highly detailed prevalence reports for autism based on their findings every 2-4 years. The most recent of these reports was released in 2014 and documented findings from a sample of 363,749 children (roughly 9% of the total population of 8-year olds in the United States) that was collected across 11 different sites in 2010 [20]. Based on the findings of this report, the overall prevalence of ASD (based on DSM IV-TR diagnostic criteria for autistic disorder, Asperger's syndrome, or PDD-NOS) for children aged 8 years in the United States is 147 per 10,000 (or 1 in 68). That figure was 29% higher than their previous estimate based on data from 2008 [22], 64% higher than their estimate based on data from 2006 [23], and a staggering 123% higher than their estimate based on data from 2002 [24]. In combination with earlier prevalence studies, the findings produced by the CDC suggest a continuous and exponential growth pattern in ASD prevalence over the past 4-5 decades (Fig. 25.1). This trend of dramatic increases in the reported prevalence of ASD has led some to the conclusion that these findings are evidence of an 'autism

epidemic' and, as a result, has encouraged increases in research focused on identifying potential environmental factors to help explain this phenomenon. However, a majority of practitioners and researchers within the ASD field have issued caution against interpreting these findings in this manner and instead suggested that the majority of these increases can be explained by a combination of alternative factors, including: broadening of diagnostic.

frequently discrepancy between the current high ASD among samples of children and the generally served prevalence rates in adults. For some, this discrepancy served as support to the idea that there has been, in fact, an ASD incidence in recent decades. In contrast to this, however, it is becoming more apparent that these discrepancies are only the result of less accurate diagnostic and identification practices in the past [25, 26]. Providing support to these beliefs, and perhaps one of the more compelling findings in opposition to the 'autism theory, a recent large community survey estimated that prevalence rates of ASD for adults in the UK should be roughly 1.1% (initially the same as those found in samples of children), but many of these adults are currently misdiagnosed or completely unidentified. Regardless of whether there has been a true increase in incidence of ASD or not, recent reports suggest that, of all mental disorders, ASD is the leading cause of disability in children below 5 years of age. Prevalence is estimated somewhere between 1% and 2%, or 52 million children [28] and a roughly equivalent proportion of adults [27] across the globe.

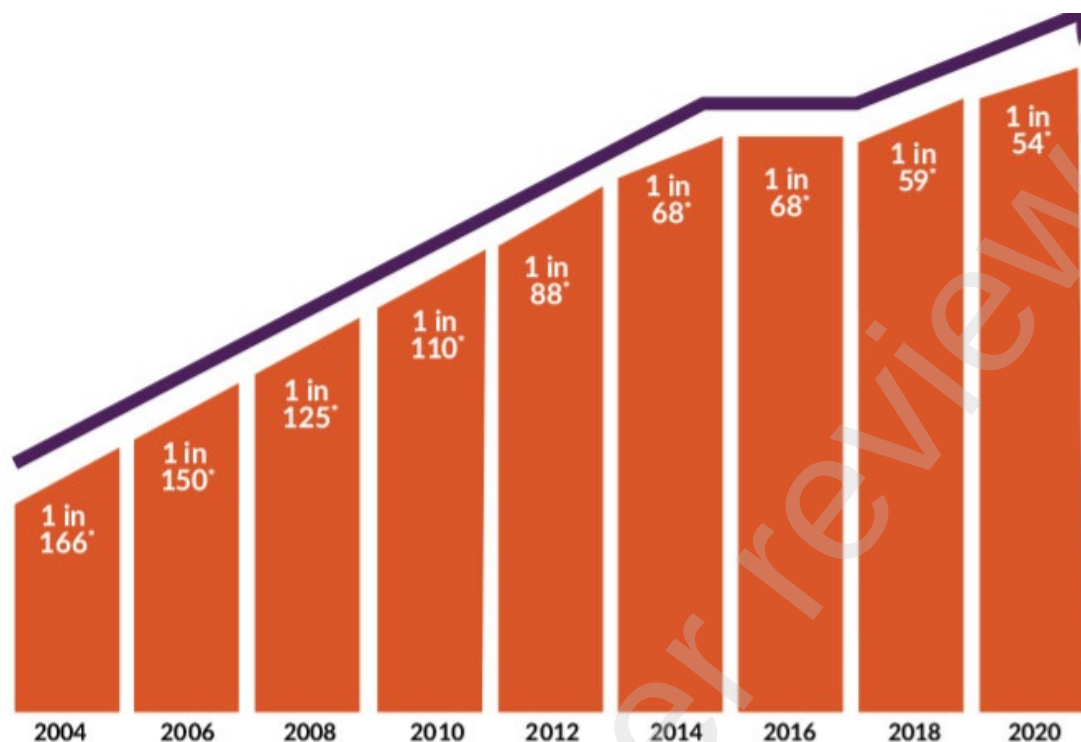


Fig 1- Centers for Disease Control (CDC) prevalence estimates of Autism spectrum disorder.

Autism Statistics and Demographic data.

One thing which has remained consistent since the earliest epidemiological studies of ASD is a From Kanner's first cohort in 1943 [4] to the first reported prevalence study in 1966 [17], and all the way through to the most recent CDC report released in 2014 [20], autism has been notably more prevalent among males than females, with estimated ratios of 2.5-5:1 [20,30]. A number of different hypotheses have been proposed for the presence of this difference. Threescore ascertainment and altered phenotypical expression in females, resulting in a with underdiagnosed [31, 323, Increased susceptibility of males due to elevated prenatal and exportable testosterone levels (331, as the existence of a female protective effect requiring a greater setio logic load in girls before

ASD Behavioural symptoms will manifest [34] At the moment, however, definitive explanation for why ASD is more prevalent in males than females has emerged. Based on current estimates (20), the male female gender ratio for ASD 4.5-1. with a light increase in this discrepancy when specifically examining individuals without co-occurring intellectual disability (49:11. Unlike gender ratios, one aspect of the demographics of individual with ASD that has changed over the years is intellectual ability. In particular, over the last decade or so, there have been notable increases in individuals identified with ASD with average or above-average intellectual ability. Likely a result of improved awareness and detection of the often subtler Behavioural manifestations of ASD in this more intellectually able cohort, recent findings suggested that 46% of children with an ASD diagnosis have IQ scores in the average to above-average range (85), 23% in the borderline range (IQ-71-85), and 31% considered to be in the range of intellectual disability (IQ 570). Finally, estimated prevalence rates of ASD tend to be higher among white/Caucasian children, compared to other ethnic/racial groups [20]. However, much of this variation can be accounted for by the significantly higher rates among white children of ASD diagnoses without co-occurring intellectual disability (a diagnosis that can be more difficult to recognize) and thus is likely more suggestive of the influence of socio-economic variables on the quality and quantity of diagnostic and treatment services available to families than of a true discrepancy of ASD prevalence between ethnic/racial groups [35].

1 in 36 kids identified with autism



241%
higher than baseline
stats in 2000

Although autism can be diagnosed before a child reaches the age of two, most kids are diagnosed only after the age of

4



BOYS are 4X more likely to be diagnosed than **GIRLS**

40% are nonverbal

44% Have average or above average intellectual ability

31% have an intellectual disability

Some areas have higher than average prevalence rates:

California 1/26
South Korea 1/38

36.5% of autism caregivers use ABA

20% of caregivers give CBD to autistic kids

Fig 2 - AUTISM QUICK STATS

Clinical features and diagnostic criteria

Although there is a general consensus that there is a strong gen-etic and biological basis to ASD, currently no definitive biological,

Neurological, or genetic markers for this disorder have been identified As a result, clinical definitions of ASD remain fundament

tally Behavioural and historical, reliant on observations (by the clinician or caregiver) of characteristic problems of social interaction, communication, and unusual behaviors which are of early onset (hence the importance of history) Complicating the task of diagnosis (and probably consistent with emerging genetic re-search (36)), it is clear that a broader spectrum of condition(s) are involved (37). For the past two decades, the approach of ICD-10 (4) and DSM IV [2, 3] have been generally concurrent, and partly as a result of this international diagnostic consistency, there has been an explode of knowledge and research. This has recently changed with the rather different approach adopted in DSM-5 [1] which focuses on a rather narrower view of autism, in the more classic sense of Kanner's autism, resulting in many more cognitively able or less prototypical cases being excluded from an ASD diagnosis (38). Further complicating the matter are the 'grandfather' clauses (adopted, in part, as a result of criticism of the increased diagnostic stringency) that allow older cases with 'well-established' DSM-IV diagnoses to retain their diagnosis. Current diagnostic criteria for ASD, as defined by DSM-5, necessitate the presentation of impairments in the following two areas [1]:

1. Persistent deficits in social interaction and social communication (including impaired social-emotional reciprocity, deficits in non-verbal social communicative behaviour, and impaired ability to develop, maintain, and understand age-appropriate social relationships).
2. Restricted and repetitive patterns of activities, behaviors, or interests (including repetitive or stereotyped motor movements, speech, or use of objects; inflexibility with regard to routine adherence, insistence on consistency, or ritualized patterns of behaviour; highly restricted interests that are atypical in focus or

intensity; and hypo- or hyper-reactivity to sensory input or atypical interest in sensory features of their environment).

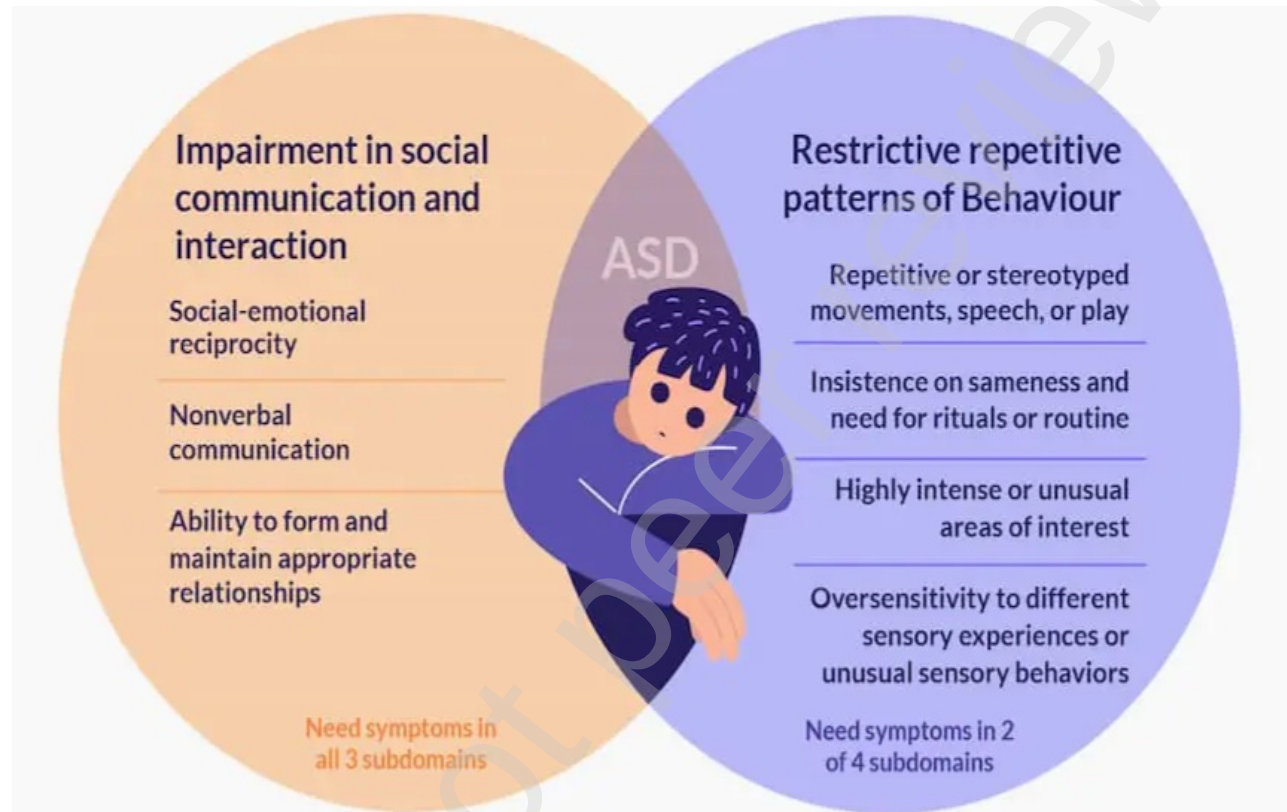


Fig 3 - ASD main areas of impairment

For a diagnosis to be received, the presence of these symptoms must be present during a child's 'early developmental period and result in clinically significant impairment in some critical area of functioning (for example, occupational, social, academic). The first of these requirements, however, includes the caveat that explicit symptoms of impairment may not become fully apparent until in later stages of life when social demands surpass an individual's limited abilities [1].

The DSM-5 approach has been extensively critiqued [38], and a recent meta-analysis has confirmed the consensus that it represents a move to a more stringent diagnostic concept [39], some what undercutting the significance of the name change from the previous category of 'Pervasive developmental disorder' to 'Autism spectrum disorder. Although perhaps intended to provide diagnostic option to those individuals not captured by this more stringent diagnostic approach, the inclusion of a new communication. disorder-social communication disorder-was introduced in DSM-5. However, the introduction of social communication dis-order presents with its own issues, given the very small amount of work supporting such a concept and the complexities involved in its application [40, 41).

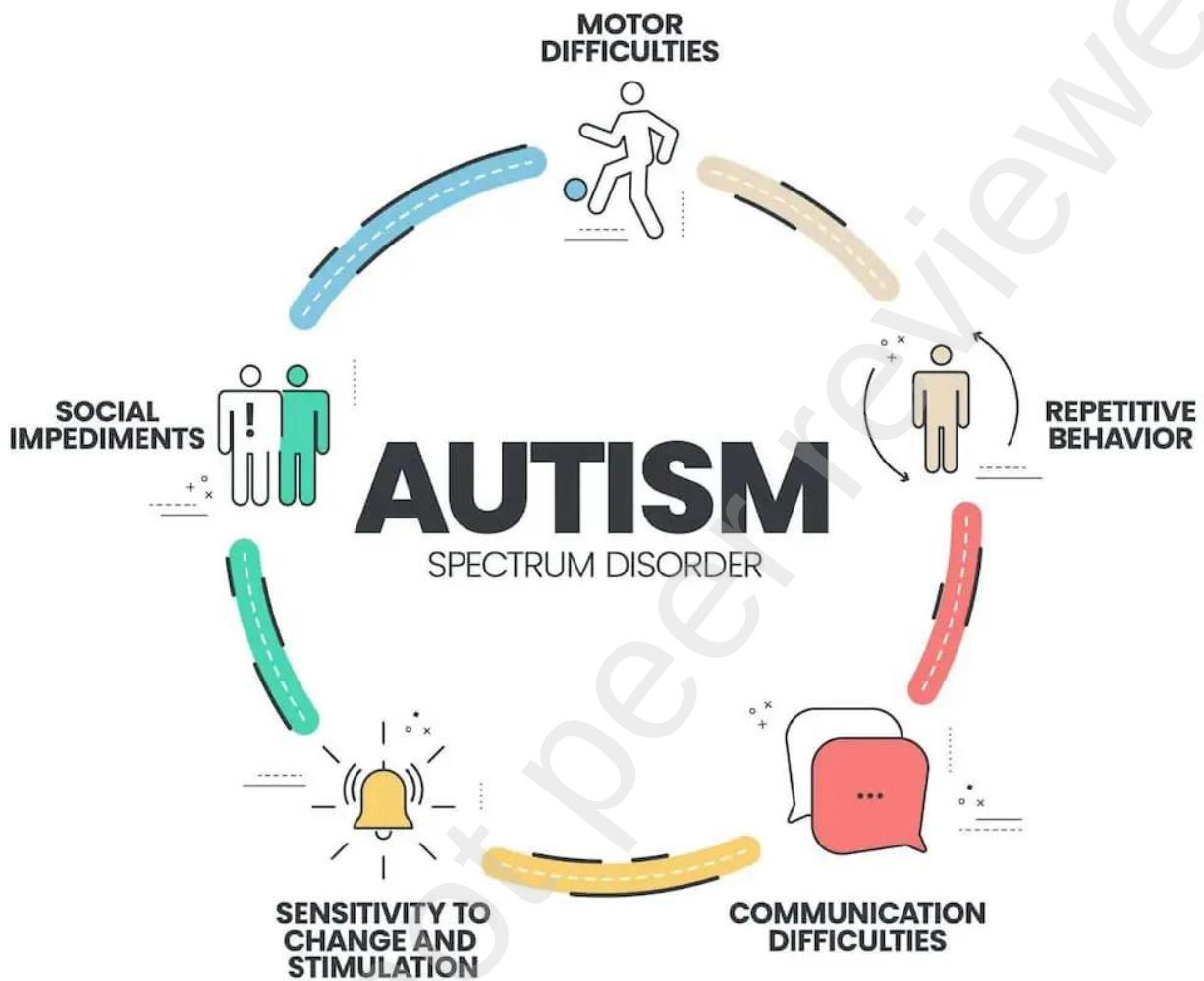


Fig 4 - Symptoms of Autism spectrum disorder

Differential diagnosis

ASD must be differentiated from other developmental disorders most frequently from intellectual disability (which co-occurs in

disorders. In intellectual disability without ASD, developmental skills (including social ones) are generally relatively evenly roughly 30% of cases of ASD [20]) and language/communication marked areas of strength (non-verbal abilities) and weakness (verbal oped (or impaired), as compared to the more typical pattern of can be most complicated in individuals with severe or profound in. and social abilities) seen in ASD. The tasks of differential diagnosis intellectual disabilities, as differences in relative areas of strength and weakness can be less discernible at these levels of cognitive impair-ment. Conversely, for more cognitively able individuals, such as those with a DSM-IV diagnosis of Asperger's syndrome, receiving task due to the grandfather clause component of DSM-5 [1]. In a new diagnosis of ASD may become an unnecessarily complicated communication disorders, social vulnerabilities are usually not par prominent and the desire to communicate often is clearly present. The new diagnosis of social communication disorder allows for a diagnosis when social and communication problems (but not restrictive behaviour) are present, but not as severe as in the more classic ASD [41]. Some individuals who go on, in later adolescence or young adult-hood, to develop schizophrenia may exhibit social oddities earlier in life, similar to, but not typically as severe as, those seen in ASD, Occasionally, severe social anxiety problems may initially be con-fused with ASD; however, these cases will generally not present with the restricted and repetitive behavior's and interests found in ASD. Conversely, the unusual (often due to intensity or repetitive nature) behavior's or interests of individuals with obsessive-compulsive disorder may be taken to suggest ASD; however, in these cases, the social and language communication skills will be preserved. Sometimes, in cases of severe neglect, a range of symptoms susuggestive ASD may be present; however, in

these cases, a history of neglect will often be observed and the social vulnerabilities will begin to abate once appropriate care is provided [42].

Key differentiating feature(s)	Autism spectrum disorder	Other conditions
Regression in development ¹⁷	An area of some controversy in ASD; further study is necessary	Typical of Rett syndrome May also occur in other conditions, such as childhood-onset schizophrenia ¹⁷
Joint attention ¹⁷ (pointing for interest, showing objects, looking at people, ³¹ spontaneously seeking to share enjoyment ¹⁰) Social drive ¹⁵	Impaired in ASD	Intact in childhood language disorders ¹⁷ (which are more common than ASD ¹⁵), although social deficits may be present due to communication difficulties secondary to language disorder
Social-communication deficits ¹⁷	More prominent in ASD	Less prominent in obsessive-compulsive disorder
Delusions and hallucinations ¹⁷	Absent in ASD	Present in childhood-onset schizophrenia
Social smiling, sharing, showing, offering (social reciprocity), nonverbal gestures ⁴³	Impaired significantly in ASD	Less impaired in Fragile X syndrome ⁴³
Repetitive behaviors ⁶	Present in ASD	Not characteristic of social pragmatic communication disorder ⁶

ASD: autism spectrum disorder

Fig 5 - Differentiating autism spectrum disorder from other conditions

Concerns and criticisms of DSM-5 classification of autism spectrum disorders

Both Kanner's original description of early infantile autism (5,7 and Asperger's description of autistic personality disorder [6] have been somewhat modified over time. For Kanner's autism, Rutter's 1978 synthesis of subsequent work [43] focused on both social and

language/communication deficits of a specific kind (not just explained by overall developmental delay) associated with problems in dealing with change and 'insistence on sameness'. Similarly, Wing's 1981 paper [15] introduced, in large part, Asperger's clinical reports to the English-speaking world, but also modified and broadened it in important ways in her definition of Asperger's syndrome [44]. Kanner's autism was officially recognized in DSM-III [14] and, up until the release of DSM-5 [1], maintained this traditional grouping of diagnostic features/criteria. Of note was the inclusion of other disorders with autism in the PDD category in DSM-IV [2,3] including Asperger's syndrome, and the convergence of ICD-10 [4] and DSM-IV definitions of these categories. the DSM-IV CD-10 definition proved quite flexible, with many possible combinations of criteria resulting in a diagnosis.

Several decisions were made at a programmatic level (for example, A number of overall changes impacted the DSM-5 review process, including having an entire child section, to change approaches to diagnosis, which impacted the process. A further consideration was the decision to use categorical classifications and not otherwise specified conditions), rather than to provide the new definitions primarily rely on data from commonly utilized ASD diagnostic tools for disorder criteria. Early on, the decision was made to move to a new diagnostic label- autism spectrum disorder (although the singular term) and avoid finer-grained classifications like Asperger's syndrome, etc. The use of modified codes and measures of severity were incorporated and were meant to provide the important information otherwise lost with the unification of previously distinct disorders under a singular label. The decision to eliminate Asperger's syndrome was most controversial, given that despite continued disagreements about the best diagnostic approach, research on that condition had

increased significantly and meta-analyses were just beginning to appear that showed important Inference's from more classic Kanner's autism, which can have critical impacts on effective treatment options and strategies (45, 46) Factor analysis of an extensive data set using measures of both current functioning and past history was utilized to come to the decision adopt a two-factor approach for the ASD diagnostic criteria in DSM-5. The two factors, determined by this analysis, included one set of criteria (that is, all four features must be present) focused on social communication problems, and a second set of polytheism criteria (that is, two of four features must be present) focused on repetitive, or atypical interests and behaviour. Post hoc analyses of the data suggested this to be a well-functioning criteria set. However, some problems have quickly become evident. In the first place, the combinations of criteria that resulted in a diagnosis of ASD were drastically reduced. Secondly, the inclusion of a new diagnostic feature associated with hypo- and hypersensitivity to sensory put did not appear well justified, as a similar feature had produced little predictive value in DSM-IV field trials [47]. In the latter, a factor analysis had shown that two-, three-, or five-factor solutions worked reasonably well (the two-factor solution paralleling the DSM-5 approach, and the five-factor solution providing additional sub-factors based on the repetitive behaviour category) [47]. It should be noted, however, that interpretations of factor analyses can be particularly susceptible to some significant issues, particularly if important data are not included and thus are unable to influence the factors being produced (see [48] for a discussion of the uses and misuses of factor analysis). Some recent studies have now questioned aspects of the factor solution proposed for DSM-5 [49, 50]. As a practical matter, it was quickly apparent that the approach adopted would disqualify many

individuals from a diagnosis-particularly those who were younger or more cognitively able (see [151] for a recent meta-analysis). As a result, a decision was made to 'grandfather' in those cases with well-established previous diagnoses, thus resulting in both DSM-IV and DSM-5 diagnostic categories being in effect simultaneously. Some aspects of the approach adopted by DSM-5 are welcome. In fact, the current diagnostic definition is, in many ways, more consistent with Kanner's classic definition of autism than the broader autism spectrum adopted in DSM-IV, which has a considerable body of support from both genetic and psychological research [38]. As noted previously, the inclusion of a new communication disorder 'social communication disorder' was not well justified and, as a practical matter, may not succeed in its purpose of capturing those individuals who will fall outside the more stringent DSM-5 diagnostic criteria for ASD, as it was not developed to be convergent with previous diagnoses of Asperger's syndrome or PDD-NOS.

Expectations for ICD-11

The eleventh edition of the WHO's International Classification of Diseases (ICD-11) major focus of this effort has been the desire to make the manual (or at least the clinical version) more user-friendly. There has been some attempt to encourage the use of a standard template for clinical decisions and diagnostic guidelines [52]. On balance, it likely will have areas of convergence with DSM-5, with some potential differences as well. A draft version is available, and efforts to evaluate it are now under way [53]. It seems likely that the ICD-11 diagnosis will certainly include individuals with ASD in DSM-5; at this point, the questions centre more around the approach to grouping certain conditions, for example, cases with regression or the more prototypical Asperger's phenotype, and whether the 'grandfathering' rule of DSM-5 will also be adopted.

It should be noted that, in addition to official systems like ICD and DSM, adoption of the Research Domain Criteria (RDOC) approach, proposed by the National Institute of Mental Health, attempts to cut across traditional definitions and focuses on domains of neurobiology and behaviour [54]. The RDOC approaches are organized around several domains of functioning and various approaches of units of study [55]. The latter can range from genes to behaviour to self-report. This approach has been extensively critiqued [56, 57] but has stimulated new lines of enquiry in fields such as co-occurring conditions in traditional approaches. As such, it may have some utility for clinical and treatment studies in which the focus or target behaviour(s) cut across transitional diagnostic categories [54, 58, 59].

Conclusion:

Autism Spectrum Disorder (ASD) is a complex neurodevelopmental condition characterized by persistent challenges with social communication and interaction, alongside restricted interests or repetitive behaviors. While there is no cure, early detection and tailored interventions can significantly improve outcomes and enable individuals with ASD to lead fulfilling lives. Research continues to unravel the interplay of genetic and environmental factors in ASD's development and variability, but significant work is still needed to understand its full scope and develop more effective treatments and services.

Footnotes:

- **Conflict of Interests:** The author declare that there is no conflict of interest regarding the publication of this article.
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